



Project Datasheet

Project Title	AI & IOT Integrated Wind Tree For Optimized Micro Wind Energy
Batch Number	01
Domain	AI Applications
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Mentor Signature	

Abstract

This project proposes the development of an *AI and IoT-enabled Wind Tree* system that generates *micro wind energy* using *recycled and reusable materials*. The goal is to create a sustainable and smart energy solution suitable for urban and residential environments where traditional wind turbines may not be feasible. The design mimics the shape of a tree, with multiple small wind turbines mounted on branches, each capable of capturing low-speed wind to generate power.

The structure is built using discarded materials such as old fan motors, PVC pipes, and used batteries. Real-time environmental and performance data is collected through sensors connected to an *ESP32 microcontroller*, which sends the information to a cloud-based or mobile dashboard using IoT technology. An embedded **AI model* analyzes wind conditions and power output to optimize turbine performance, predict energy generation, and ensure system efficiency through intelligent control.

Expected outcomes include effective energy generation in low-wind environments, improved performance through AI-based optimization, and continuous monitoring via IoT. The project demonstrates a practical blend of *renewable energy*, **smart technology*, and **eco-friendly design* using accessible and low-cost materials.

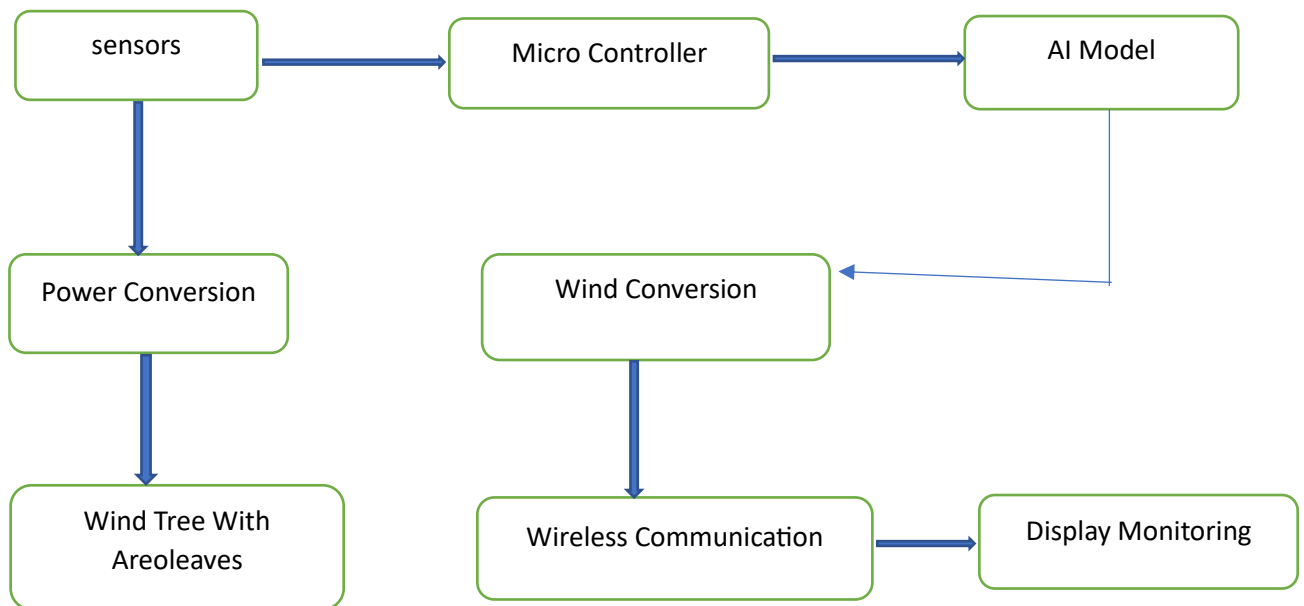
This innovation highlights how emerging technologies like AI and IoT can be applied to support sustainable energy practices and promote environmental responsibility in real-world applications.

Keywords: Micro wind energy, Wind Tree, IoT, Artificial Intelligence, Recycled materials, Sustainable technology, Renewable energy

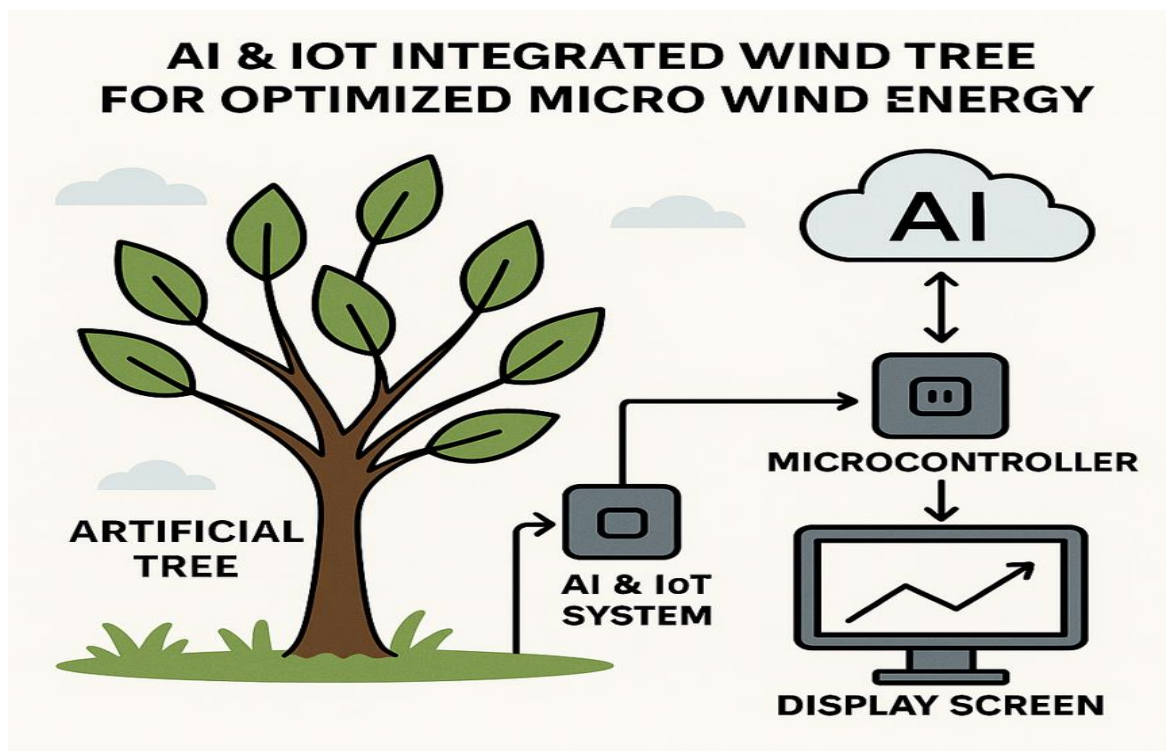
Objective

- The Main motive of this project is to design and fabricate a low-cost Wind Tree prototype that harnesses wind energy using multiple aero-leaves arranged in a tree-like structure.
- To integrate IoT sensors for real-time monitoring of wind speed, turbine RPM, and energy generation statistics.

Block Diagram



3D Diagram of the proposed prototype



Functional layout of the project: